Environmental Science and Pollution Research (2022) 29:37527–37542
https://doi.org/10.1007/s11356-021-18163-1

RESEARCH ARTICLE

Multi-drug resistance, integron and transposon-mediated gene transfer in heterotrophic bacteria from *Penaeus vannamei* and its culture environment

Ranjit Kumar Nadella[1] · Satyen Kumar Panda[2] · Madhusudana Rao Badireddy[3] · Pani Prasad Kurcheti[4] · Ram Prakash Raman[4] · Mukteswar Prasad Mothadaka[1]

Received: 25 August 2021 / Accepted: 13 December 2021 / Published online: 23 January 2022

Abstract

Multi-drug resistance (MDR) in bacteria is regarded as an emerging pollutant in different food production avenues including aquaculture. One hundred and sixty out of 2304 bacterial isolates from shrimp farm samples ($n = 192$) of Andhra Pradesh, India, were MDR. Based on biochemical identification and 16S rRNA sequencing, they were grouped into 35 bacterial species with the predominance of *Vibrio parahaemolyticus* (12.5%). The MDR isolates showed highest resistance toward oxytetracycline (89%) with more than 0.2 MAR (multiple antibiotic resistance), demonstrates a high-risk source. The most prevalent antibiotic-resistance gene (ARG) and mobile genetic element (MGE) detected were *tetA* (47.5%) and *int1* (46.2%), respectively. In conjugation experiments, overall transfer frequency was found to be in the range of 1.1×10^{-9} to 1.8×10^{-3} with the transconjugants harbouring ARGs and MGEs. This study exposed the wide distribution of MDR bacteria in shrimp and its environment, which can further aggravate the already raised concerns of antibiotic residues in the absence of proper mitigation measures.

Keywords Antibiotic resistance genes (ARGs) · Conjugation · Integron · Multi-drug resistant (MDR) · Mobile genetic elements (MGEs) · Shrimp farms · Transposon

Introduction

Antibiotics are commonly used in different food production avenues for treating or preventing diseases in humans and rearing animals (Qiao et al. 2018). In aquaculture production systems, especially in the case of intensive culture practices, antimicrobials are applied for prophylactic and therapeutic purposes for disease control (Rakesh et al. 2018), which is seen more often in shrimp farming (Thornber et al. 2020) leading to persistence of antibiotics in rearing environments (Preena et al. 2020). Globally, the white-leg shrimp (*Penaeus vannamei*) is the major and economically sustainable variety of cultured shrimp contributing 4.9 million tonnes (52.9%) of the whole crustacean production in the year 2018 (FAO 2020). Similarly, during 2019–2020, *P. vannamei*'s share in total farmed shrimp production from India and Andhra Pradesh was ~ 711,000 and 510,000 tonnes, respectively (MPEDA 2020).

Antibiotics belonging to various classes viz., tetracyclines, sulphonamides, quinolones, macrolides and phenicols have been employed in shrimp aquaculture

Responsible editor: Diane Purchase

Mukteswar Prasad Mothadaka
prasadmm@hotmail.com

1 MFB Division, ICAR-Central Institute of Fisheries Technology, Willingdon Island, Matsyapuri P.O., Cochin 682029, Kerala, India

2 QAM Division, ICAR-Central Institute of Fisheries Technology, Willingdon Island, Matsyapuri P.O, Cochin 682029, Kerala, India

3 Visakhapatnam Research Centre, ICARCentral Institute of Fisheries Technology, Visakhapatnam 530003, Andhra Pradesh, India

4 Aquatic Environment and Health Management Division, ICAR-Central Institute of Fisheries Education, Versova, Mumbai 400061, Maharashtra, India

(Bermúdez-Almada and Espinosa-Plascencia 2012). As a result, antibiotic residues have been noticed in farmed shrimp (Guo et al. 2018) which led to the rejection of numerous consignments of shrimp exports (Rao and Prasad 2015; Panda and Ravishankar 2018). Antibiotics are generally applied for control of infections caused by pathogenic microorganisms. Indiscriminate use and persistent presence of antibiotic residues also contribute to the acquiring of antimicrobial resistance (AMR) in the bacteria due to selective pressure (Cabello et al. 2013). Multi-drug resistance (MDR) is a phenomenon in which bacteria acquire resistance to minimum of one antibiotic in three or more classes of antibiotics (Magiorakos et al. 2012) due to accumulation of the genes that code for multiple drugs on resistance (R) plasmids (Nikaido 2009). Exposure of bacteria to antibiotics can facilitate the development and spread of antibiotic resistance genes (ARGs) that pose potent hazard to human health and the environment (Huang et al. 2015; Murugadas et al. 2019). Besides, aquatic environments have been considered as a pool for MDR bacteria and genes (Marti et al. 2014). Hence, the presence of ARGs may be considered as emerging contaminants in various environs (Sanderson et al. 2016).

In shrimp aquaculture, numerous studies are carried on isolation and characterization on MDR pattern in pathogenic bacteria (Kathleen et al. 2016; Teng et al. 2017; Pham et al. 2018; Dhayanath et al. 2019; Narayanan et al. 2020). However, information on the occurrence of MDR in bacteria, especially from the culture environs of *P. vannamei*, is scant. Similarly, several authors reported the occurrence of ARGs in pathogen-specific culturable bacteria from shrimp culture environment (Pham et al. 2018; Zhou et al. 2019; Girijan et al. 2020) and also in commensal bacteria (Su et al. 2017; Hong et al. 2018; Liu et al. 2019; Zhou et al. 2020). In aquatic environs, the part played by the mobile genetic elements (MGEs) is substantial in the transfer of ARGs between bacteria by horizontal gene transfer (HGT) (Sharma et al. 2016; Zhang et al. 2020). As there is high proportion of HGT among the commensal bacteria by MGEs, information on the incidence of ARGs in bacterial isolates helps in improved understanding of dissimilar tools connected with the diffusion of drug resistance (von Wintersdorff et al. 2016). Thus, it becomes essential to assess the prevalence of ARGs and their related MGEs in shrimp farming environment to pinpoint the AMR menace (Cabello et al. 2013). In this study, we report the occurrence and characterization of MDR bacteria, ARGs and MGEs harboured by the heterotrophic culturable bacteria isolated from shrimp (*P. vannamei*) cultured ponds of Andhra Pradesh, India.

Materials and methods

Isolation and identification of bacterial isolates

Heterotrophic bacteria were isolated from 192 samples (farm sediment (*n*-60), pond water (*n*-60), cultured shrimp (*n*-60) and reservoir water (*n*-12)) collected for a period of 2 years from January, 2017 to December, 2018 from *P. vannamei* culture ponds at 12 different geographical locations (L1 to L12) from four districts namely Srikakulam (north coastal region), East Godavari and West Godavari (central coastal regions) and Sri Potti Sriramulu Nellore (south coastal region) of Andhra Pradesh state in India (Fig. 1). From respective geographical locations, five ponds were selected within a distance of 2–6 km between any two ponds. The ponds were selected based on difference in stocking densities, area, days of culture and diverse culture practices encompassing the entire region of Andhra Pradesh. From each pond, sediment (S), shrimp (Sh) and water (W) samples were drawn aseptically and immediately transported to the laboratory under iced conditions. Standard methods were employed in the isolation of bacteria (ISO 4833–2: 2013; ISO 6222: 1999) with minor alteration of using Marine Agar. Identification of the isolated bacteria up to genus level was carried out by employing conventional tests (morphological and biochemical) (Brenner et al. 2005), and species level identification was done by performing additional tests for each species by referring to earlier studies (Table S1a–S1d, Supporting Information).

Assessment of antibiotic susceptibility

Antibiotic susceptibility of bacterial isolates was performed by disc diffusion method (CLSI 2020) against five groups of antibiotics viz., quinolones (ciprofloxacin), phenicols (chloramphenicol), folate pathway inhibitors (co-trimoxazole), macrolides (erythromycin) and tetracyclines (oxytetracycline). *Escherichia coli* (ATCC 25922) and *Staphylococcus aureus* (ATCC 29213) were used as quality control cultures for comparison. The antibiotic discs (Himedia, Mumbai) used in the study were: oxytetracycline — 30 µg, ciprofloxacin — 5 µg, co-trimoxazole — 10 µg, chloramphenicol — 30 µg and erythromycin — 15 µg.

DNA extraction and 16S rRNA gene sequencing

Crude DNA extraction from MDR isolates was carried out by heat lysis method (Capkin et al. 2015). The MDR

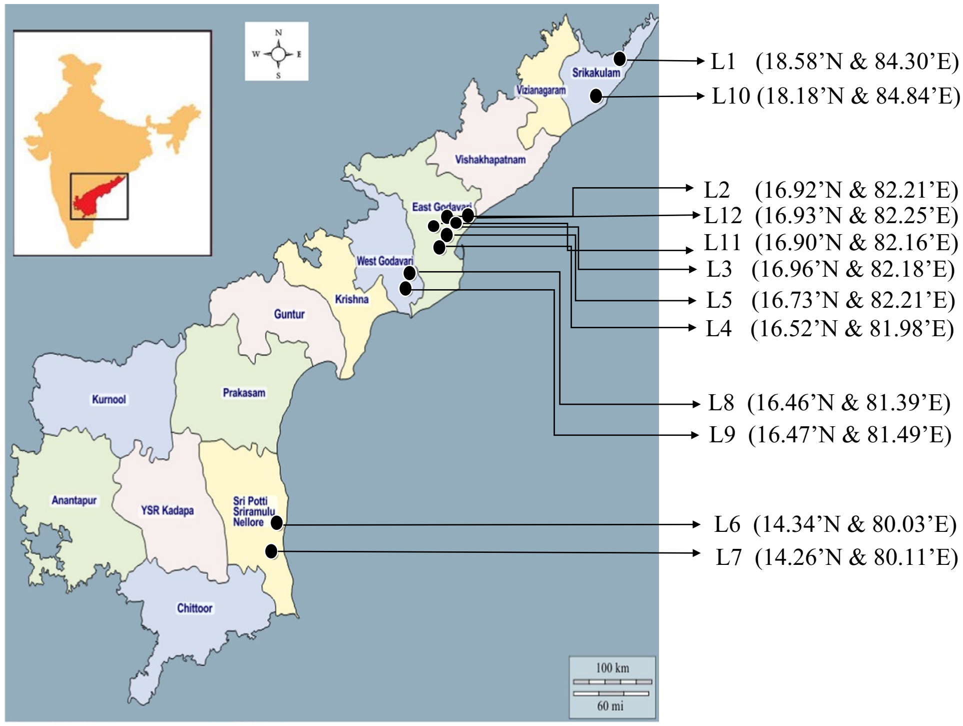

Fig. 1 Locations of shrimp ponds in different districts of Andhra Pradesh, India from where the farm samples were collected

isolates were randomly selected for PCR, targeting 16S rRNA region of the bacteria employing universal primers (27F and 1492R) (Weisburg et al. 1991). The purified PCR products were Sanger sequenced at Scigenome Pvt. Ltd. Cochin, India by outsourcing.

PCR amplification of ARGs, integron and transposons

All phenotypically multi-drug resistant isolates were tested by PCR assay for harbouring of ARGs and MGEs by employing specific primers and amplifying conditions as mentioned in Table S2 (supplementary data). The ARGs screened were *tetA*, *tetK*, *tetM*, *tetL*, *tetS* for tetracycline; *qnrA*, *qnrD* for ciprofloxacin; *sul1*, *sul2* for sulphonamide; *ermB*, *ermC* for erythromycin and *cat1*, *catII*, *florR* for chloramphenicol in drug-resistant isolates. Similarly, multiplex PCR was carried out to detect integrase gene carried by class 1 and 2 integrons, employing specific primers. Transposons *Tn1721*, *Tn1545* and *Tn917* were also screened by using specific primers. The sequences of the primers and the conditions employed for PCR assay of each ARG and MGE are shown in Table S2 (Supporting Information). In each cycle, a positive control (plasmid DNA from *Escherichia coli* ATCC 43,888) and negative control (nuclease free water) were included. The PCR reaction mixture was prepared with a composition of 12.5 µL of master mix (Origin, India), template DNA, 0.2 pmol/µL primers (forward and reverse) and the volume raised to 25 µL with the water free from nuclease.

Conjugation and Minimum Inhibitory Concentration (MIC) studies

Conjugation experiments were carried out by using recipient cells (plasmid-free *Escherichia coli* DH5α as recipient which was resistant to nalidixic acid (NA^R) (20 µg mL^{-1})) and the donor cells (fifteen MDR bacterial isolates resistant to all the five tested antibiotics) (Suhartono and Savin 2016). The bacterial isolates were grown overnight with shaking (200 rpm) at 35 °C in Luria broth (LB). After incubation, the donor and recipient bacterial cells were mixed in the ratio of 1:3 and allowed to conjugate overnight at 35 °C. Later, the grown cultures were serially diluted up to 10^{-6} dilution, and 0.1 ml from each dilution was plated on Mueller–Hinton (MH) agar supplemented with oxytetracycline (8 µg mL^{-1}), sulfamethoxazole (8 µg mL^{-1}), ciprofloxacin (0·25 µg mL^{-1}), erythromycin (0.1 µg mL^{-1}), chloramphenicol (25 µg mL^{-1}) for donor cells and nalidixic acid (20 µg mL^{-1}) for recipient *E. coli* DH5α cells. Aliquots were also plated on MH agar added with nalidixic acid (20 µg mL^{-1}) as a control for the recipient bacterial cells. The plating procedures were done in replica, and the petri dishes were incubated at 35 °C for 24–48 h. Further, enumeration was carried out on colony counter, and transfer frequency was calculated as per the formula based on the ratio of number of transconjugant cells (cfu) to number of recipient cells (cfu) multiplied with dilution factor (Leungtongkam et al. 2018). From the transconjugant cells, plasmid was isolated by using GeneJET plasmid Miniprep kit (Thermoscientific, USA) as per the maker's directions and screened for the above mentioned ARGs and MGEs. Minimum inhibitory concentration (MIC) test for the transconjugants was carried out as per the standard procedure (broth dilution) by employing MH broth (CLSI 2020). Reference cultures *Escherichia coli* (ATCC 25922) and *Staphylococcus aureus* (ATCC 29213) were employed in parallel as control strains.

Calculation of MAR index

Krumperman's formula was used to estimate multiple antibiotic resistance (MAR) index of the MDR isolates based on method *a*/*b*, where "*a*" denotes the tested antibiotics to which the strains were resistant, and "*b*" denotes the total number of antibiotics screened (Krumperman 1983).

Data analysis

The data were analysed by employing Microsoft excel 2010 (Microsoft Corporation, Redmond, WA) with two-factor analysis of variance (ANOVA) to test the significant differences in AMR of bacterial isolates between different locations and between different morphological groups of bacterial isolates respectively.

Results

Isolation and identification of bacterial isolates by conventional biochemical tests

Significant number of ($n=2304$) bacteria were isolated from the pond samples of shrimp aquaculture collected at diverse geographical sites (L1 to L12) of Andhra Pradesh, India (Table 1). The dominant bacteria were *Vibrio* sp. ($n=888$) followed by *Bacillus* sp. ($n=456$). Similarly, highest antibiotic resistance and MDR were observed in *Vibrio* sp. ($n=545$; 64) followed by *Bacillus* sp. ($n=326$; 34). Based on the morphological and conventional biochemical tests, MDR bacteria were identified into sixteen genera in four groups namely Group A: *Bacillus* sp., *Microbacterium* sp., *Bhargavaea* sp., *Exiguobacterium* sp.; Group B: *Vibrio* sp., *Halomonas* sp., *Tenacibaculum* sp., *Pseudoalteromonas* sp., *Escherichia coli*, *Pseudomonas* sp., *Burkholderia* sp.; Group C: *Kocuria* sp., *Salinicoccus* sp., *Staphylococcus* sp., *Brachybacterium* sp., and Group D: *Acinetobacter* sp. Based on additional biochemical

Table 1 Bacterial cultures ($n=2304$) isolated from shrimp pond samples collected at different locations, antibiotic resistant and MDR isolates

Bacterial genera	Sample collection locations												Bacterial isolates		
	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	Total	Resistant	MDR
Group A: Gram positive rods															
Bacillus sp	35	41	39	32	28	41	42	33	36	45	41	43	456	236	34
Microbacterium sp	5	7	5	4	5	5	7	6	2	8	6	5	65	42	6
Bhargavaea sp	2	3	1	3	2	3	2	3	2	1	0	2	24	15	1
Exiguobacterium sp	7	5	8	7	6	8	4	7	6	8	3	8	77	48	5
Group B: Gram negative rod															
Vibrio sp	72	79	77	70	79	68	77	66	69	74	78	79	888	545	64
Halomonas sp	11	10	12	12	13	13	12	10	12	13	11	10	139	95	9
Tenacibaculum sp	10	12	13	11	16	12	9	14	12	15	13	12	149	82	6
Pseudoalteromonas sp	4	6	3	5	4	4	3	5	5	6	8	6	59	33	5
Escherichia coli	3	2	4	3	4	5	6	3	2	4	3	3	42	28	4
Pseudomonas sp	10	11	9	11	12	8	12	11	10	10	10	9	123	66	5
Burkholderia sp	1	0	1	2	0	1	2	1	0	1	2	1	12	6	3
Group C: Gram positive cocci															
Kocuria sp	2	3	3	2	2	3	0	2	2	3	0	3	25	14	2
Staphylococcus sp	9	10	11	7	12	10	8	9	10	10	11	11	118	71	6
Salinicoccus sp	4	2	5	3	4	2	2	3	3	2	3	4	37	24	3
Brachybacterium sp	1	2	1	1	2	0	1	1	1	1	1	0	12	4	2
Group D: Gram negative coccobacilli															
Acinetobacter sp	8	6	7	5	7	6	9	8	5	6	5	6	78	43	5
Total	184	199	199	178	196	189	196	182	177	207	195	202	2304	1352	160

tests, the above drug-resistant isolates were further classified into 35 species.

16S rRNA sequencing analysis

Analysis of 16S rRNA sequences of the multi-drug resistant bacterial isolates showed close similarity to 35 species belonging to 16 genera namely *Acinetobacter soli*, *Bacillus haikouensis*, *Bacillus oryzaecorticis*, *Bacillus paramycoides*, *Bacillus safensis*, *Bacillus subtilis*, *Bhargavaea beijingensis*, *Brachybacterium paraconglomeratum*, *Burkholderia cepacia*, *Cytobacillus firmus*, *Escherichia coli*, *Exiguobacterium mexicanum*, *Halobacillus litoralis*, *Halomonas ventosae*, *Kocuria flava*, *Lysinibacillus sp.*, *Mesobacillus foraminis*, *Mesobacillus jeotgali*, *Mesobacillus subterraneus*, *Microbacterium aurum*, *Microbacterium koreense*, *Pseudoalteromonas piscicida*, *Pseudomonas knackmussii*, *Pseudomonas stutzeri*, *Salinicoccus roseus*, *Staphylococcus aureus*, *Tenacibaculum aestuarii*, *Tenacibaculum lutimaris*, *Vibrio alginolyticus*, *Vibrio azureus*, *Vibrio diazotrophicus*, *Vibrio mediterranei*, *Vibrio parahaemolyticus*, *Vibrio vulnificus* and *Virgibacillus pantothenticus*. The sequences obtained were compared with the data reference isolates and later submitted in the public domain (NCBI data base) with accession numbers as mentioned in Table 2.

AMR of MDR bacterial isolates in shrimp pond samples

Data related to antibiotic resistance of multi-drug resistant bacterial isolates is summarised in Fig. 2. Among one hundred and sixty MDR isolates, highest resistance was shown by isolates towards oxytetracycline (89%) followed by erythromycin (78%) (Fig. 2a). Shrimp pond location-wise, highest antibiotic resistance was recorded for ciprofloxacin at L12 (100%), chloramphenicol at L12 (100%), co-trimoxazole at L8 (100%), erythromycin at L9, L11, L12 (100%) and oxytetracycline at L1, L2, L3, L7, L8, L9 (100%) (Fig. 2b). Morphological group-wise, Gram negative (GN) rods exhibited relatively higher resistance to ciprofloxacin (87%), chloramphenicol (44%), co-trimoxazole (66%), erythromycin (87.6%) and oxytetracycline (92.8%) (Fig. 2c). Not any noteworthy difference ($p \geq 0.05$) was seen in AMR between different locations and between different morphological bacterial groups. Bacterial species-wise, highest resistance for ciprofloxacin (100%), chloramphenicol (100%), co-trimoxazole (100%); erythromycin (100%), oxytetracycline (100%) was observed in thirteen, three, seven, fourteen and nineteen species respectively (Table 3). One of the important observations was isolates belonging to *Tenacibaculum aestuarii* and *Bacillus paramycoides* recorded 100% resistance to all the tested antibiotics.

MAR index of multi-drug resistant isolates

The data on MAR index of MDR bacterial isolates is presented in Table 4. Minimum MAR index recorded was 0.6 for all bacterial species as they exhibited resistance to minimum of three or more antibiotics of different classes, whereas maximum MAR index of 1 was recorded for *Bacillus paramycoides*, *Bacillus safensis*, *Escherichia coli*, *Halomonas ventosae*, *Pseudoalteromonas piscicida*, *Tenacibaculum lutimaris*, *Vibrio alginolyticus*, *Vibrio azureus*, *Vibrio mediterranei* and *Vibrio parahaemolyticus*. The MAR index (100%) was identified between 0.6 and 0.8 in 11 bacterial species and between 0.8 and 1 in three bacterial species.

Prevalence of ARGs and MGEs in MDR isolates

The data pertaining to prevalence of ARGs among different bacterial species isolated from shrimp pond samples is presented in Table 5. The most prevalent ARG was *tetA* (47.5%) in tetracycline group, *sul1* (26.3%) in sulphonamide group, *qnrD* (20%) in quinolone group, *catI* (19.4%) in chloramphenicol group and *ermB* in (18.8%) in macrolide group. A low percentage of prevalence was noticed for *tetL* (9.4%) midst of all the screened ARGs. Among bacterial species, *Vibrio parahaemolyticus* showed the highest number of isolates positive for *tetA* gene ($n=10$) followed by *Vibrio alginolyticus* ($n=9$). Among MGEs screened in the MDR isolates, integrons *int1* and *int2* were noticed in 46.2% and 18.7% of the isolates, respectively. Similarly, transposons *Tn1721*, *Tn917* and *Tn1545* were observed in 15.6%, 13%, 7% and 5% respectively. PCR amplification products of ARGs and MGEs prevalent in MDR isolates is shown in Fig. S1 and Fig. S2, respectively (Supporting information).

Conjugation and MIC studies

Results of conjugation studies carried out to study the transfer of ARGs between the species of bacteria are depicted in Table 6. The conjugation transfer frequency for MDR bacterial isolates was recorded in overall range of 1.8×10^{-3} (*Pseudoalteromonas piscicida*) to 1.1×10^{-9} (*Vibrio parahaemolyticus*) transconjugants per recipient cell. The conjugation transfer frequency for the tested antibiotics was found in the range of 1.3×10^{-6} (L6DW4) to 4.6×10^{-8} (L2CS9) for ciprofloxacin; 3.8×10^{-3} (L5CSh9) to 6.5×10^{-5} (L7BSh1) for chloramphenicol; 1.4×10^{-3} (L6DW4) to 2.9×10^{-5} (L5CSh9) for co-trimoxazole; 2.3×10^{-7} (L1BSh3) to 1.1×10^{-9} (L5ESh7) for erythromycin and 9.1×10^{-6} (L7DS12) to 1.2×10^{-8} (L5ESh7) for oxytetracycline. The transconjugants also harboured the ARGs such

Table 2 Accession numbers of MDR bacteria submitted in public domain (NCBI)

S no	Isolate code	Location	Sample source	Accession number	Identified as	Per cent identity with NCBI strain (%)
1	L4BSh8	L4	Shrimp	MW295538	*Acinetobacter soli*	99.8
2	L5ASh4	L5	Shrimp	MW295539	*Bacillus haikouensis*	100
3	L7DS12	L7	Sediment	MW454362	*Bacillus oryzaecorticis*	100
4	L2BW10	L2	Water	MW295533	*Bacillus paramycoides*	100
5	L3ES3	L3	Sediment	MW295625	*Bacillus safensis*	99.7
6	L6DW4	L6	Water	MW454342	*Bacillus safensis*	100
7	L8CS3	L8	Sediment	MW295585	*Bacillus subtilis*	100
8	L12AW5	L12	Water	MW295821	*Bhargavaea beijingensis*	100
9	L4ESh3	L4	Shrimp	MW295574	*Brachybacterium paraconglomeratum*	100
10	L6RW6	L6	Water	MW454119	*Brachybacterium paraconglomeratum*	100
11	L10DS4	L10	Sediment	MW295834	*Burkholderia cepacia*	100
12	L3ASh9	L3	Shrimp	MW295827	*Cytobacillus firmus*	100
13	L5CSh9	L5	Shrimp	MW295580	*Escherichia coli*	100
14	L6DSh6	L6	Shrimp	MW454085	*Escherichia coli*	98.1
15	L10CS1	L10	Sediment	MW295833	*Exiguobacterium mexicanum*	97.2
16	L7EW7	L7	Water	MW295820	*Halobacillus litoralis*	99.4
17	L1ES7	L1	Sediment	MW295579	*Halomonas ventosae*	99.4
18	L2CS7	L2	Sediment	MW454086	*Halomonas ventosae*	99.6
19	L6BW9	L6	Water	MW295835	*Kocuria flava*	99.8
20	L2AW6	L2	Water	MW295824	*Lysinibacillus sp.*	98.2
21	L11CSh5	L11	Shrimp	MW295828	*Mesobacillus foraminis*	99.1
22	L7AW10	L7	Water	MW295822	*Mesobacillus jeotgali*	94.4
23	L6EW1	L6	Shrimp	MW295830	*Mesobacillus subterraneus*	100
24	L11DSh9	L11	Shrimp	MW295542	*Microbacterium aurum*	98.2
25	L2DSh10	L2	Shrimp	MW295628	*Microbacterium koreense*	99.8
26	L2CS9	L2	Sediment	MW295588	*Pseudoalteromonas piscicida*	98.5
27	L6DS3	L6	Sediment	MW295832	*Pseudomonas knackmussii*	99.6
28	L7Bsh3	L7	Shrimp	MW295831	*Pseudomonas stutzeri*	92.9
29	L11AW2	L11	Water	MW295555	*Salinicoccus roseus*	99.8
30	L8RW5	L8	Water	MW454125	*Salinicoccus roseus*	98.4
31	L10EW2	L10	Water	MW295826	*Staphylococcus aureus*	98.5
32	L2BSh3	L2	Shrimp	MW454117	*Tenacibaculum aestuarii*	98.7
33	L1BSh3	L1	Shrimp	MW295541	*Tenacibaculum lutimaris*	99.1
34	L1ASh8	L1	Shrimp	MW454349	*Tenacibaculum lutimaris*	99.1
35	L2ESh10	L2	Shrimp	MW454353	*Vibrio alginolyticus*	100
36	L4ASh5	L4	Shrimp	MW454346	*Vibrio alginolyticus*	100
37	L9DSh4	L9	Shrimp	MW454350	*Vibrio alginolyticus*	100
38	L9ESh2	L9	Shrimp	MW295554	*Vibrio alginolyticus*	100
39	L8BSh2	L8	Shrimp	MW454113	*Vibrio alginolyticus*	99.5
40	L10ASh10	L10	Shrimp	MW454348	*Vibrio azureus*	100
41	L3CSh6	L3	Shrimp	MW295819	*Vibrio diazotrophicus*	100
42	L6BSh7	L6	Shrimp	MW454112	*Vibrio mediterranei*	100
43	L3BSh5	L3	Shrimp	MW295587	*Vibrio mediterranei*	98.9
44	L9AS7	L10	Sediment	MW454363	*Vibrio mediterranei*	99.7
45	L4ASh8	L4	Shrimp	MW454377	*Vibrio parahaemolyticus*	100
46	L4DSh3	L4	Shrimp	MW454339	*Vibrio parahaemolyticus*	99.6
47	L5ESh2	L5	Shrimp	MW454368	*Vibrio parahaemolyticus*	99.7
48	L5ESh7	L5	Shrimp	MW454369	*Vibrio parahaemolyticus*	99.8

Table 2 (continued)

S no	Isolate code	Location	Sample source	Accession number	Identified as	Per cent identity with NCBI strain (%)
49	L9BSh1	L9	Shrimp	MW295553	*Vibrio parahaemolyticus*	99.5
50	L7BSh1	L7	Shrimp	MW454115	*Vibrio parahaemolyticus*	100
51	L1AW12	L1	Water	MW295504	*Vibrio vulnificus*	99.5
52	L3DS5	L3	Sediment	MW454352	*Virgibacillus pantothenticus*	100
53	L8ESh12	L8	Shrimp	MW295537	*Virgibacillus pantothenticus*	99.8

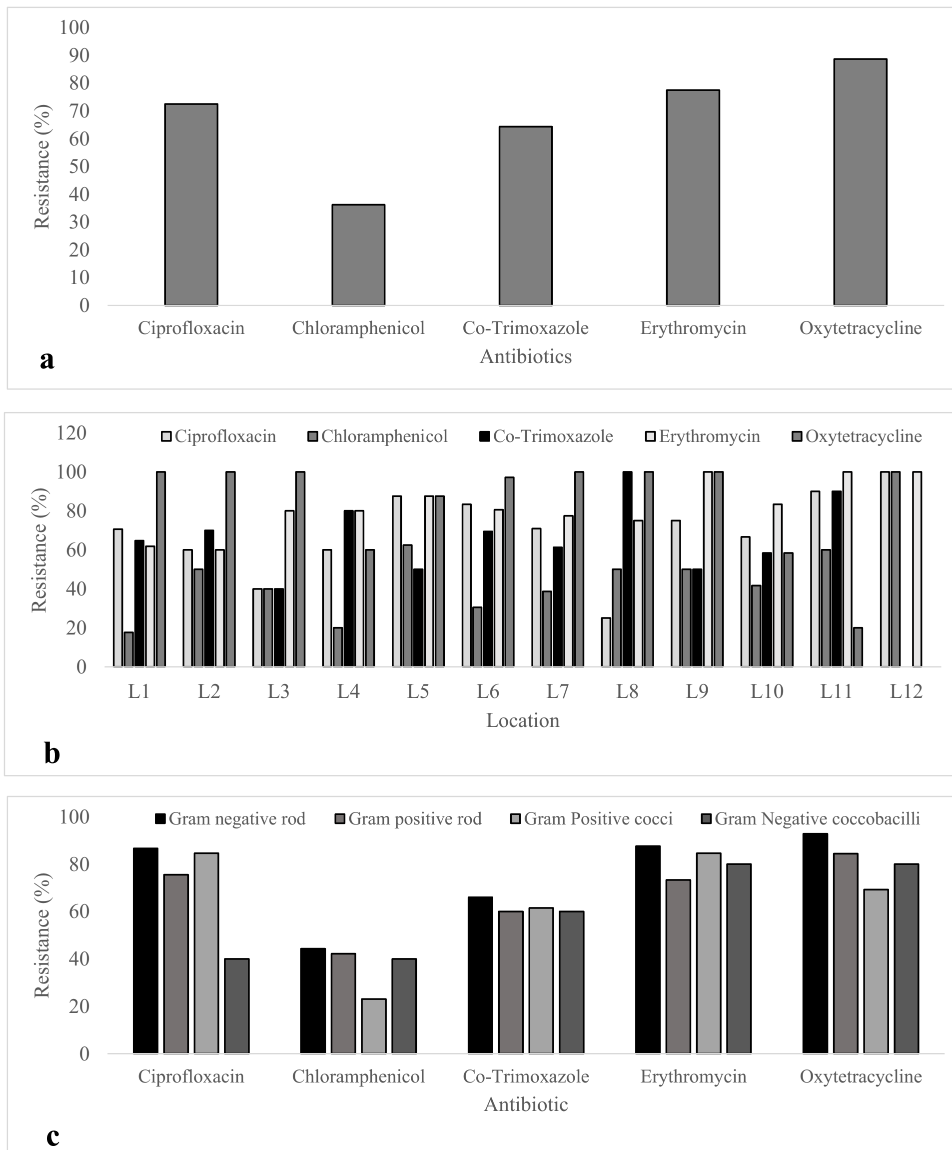

Fig. 2 Antibiotic resistance percentage of multi-drug bacterial isolates from shrimp samples. **a** between antibiotics, **b** between locations, **c** between different morphotintorial groups of bacteria

as *tetA*, *sul1*, *ermB* and *qnrA* which were acquired from the parent cultures. Similarly, the transconjugants too harboured MGEs such as *int1*, *Tn1721* and *Tn1545*. The data revealed that the minimum inhibitory concentration for the bacterial cultures was seen from 0 to 0.5 μg mL^{-1} for ciprofloxacin, 32 to 128 μg mL^{-1} for chloramphenicol, 64 to 512 μg mL^{-1} for co-trimoxazole, 32 to 128 μg mL^{-1} for erythromycin and 32 to 512 μg mL^{-1} for oxytetracycline.

Table 3 Antibiotic resistance percentage of multi-drug resistant bacteria isolated from shrimp aquaculture farms

Bacterial species	Number	Antibiotic resistance (%)				
		Ciprofloxacin	Chloramphenicol	Co-trimoxazole	Erythromycin	Oxytetracycline
Acinetobacter soli	5	40.0	40.0	80.0	100.0	100.0
Bacillus haikouensis	2	100.0	0.0	0.0	100.0	100.0
Bacillus oryzaecorticis	2	100.0	50.0	100.0	0.0	100.0
Bacillus paramycoides	2	100.0	100.0	100.0	100.0	100.0
Bacillus safensis	4	75.0	25.0	75.0	75.0	100.0
Bacillus subtilis	5	60.0	40.0	40.0	80.0	100.0
Bhargavaea beijingensis	1	100.0	100.0	0.0	100.0	0.0
Brachybacterium paraconglomeratum	2	50.0	0.0	100.0	100.0	50.0
Burkholderia cepacia	3	66.7	66.7	33.3	66.7	66.7
Cytobacillus firmus	1	0.0	0	0.0	100.0	100.0
Escherichia coli	4	100.0	50.0	50.0	100.0	100.0
Exiguobacterium mexicanum	5	100.0	60.0	40.0	60.0	80.0
Halobacillus litoralis	4	75.0	25.0	75.0	100.0	75.0
Halomonas ventosae	9	77.8	33.3	77.8	77.8	88.9
Kocuria flava	2	100.0	0.0	50.0	50.0	100.0
Lysinibacillus sp.	2	50.0	50.0	50.0	50.0	100.0
Mesobacillus foraminis	3	100.0	66.7	33.3	100.0	33.3
Mesobacillus jeotgali	3	66.7	0.0	100.0	66.7	100.0
Mesobacillus subterraneus	2	100.0	50.0	50.0	50.0	100.0
Microbacterium aurum	4	75.0	25.0	75.0	50.0	75.0
Microbacterium koreense	2	50.0	50.0	50.0	100.0	50.0
Pseudoalteromonas piscicida	5	80.0	40.0	80.0	80.0	100.0
Pseudomonas knackmussii	2	100.0	50.0	100.0	50.0	50.0
Pseudomonas stutzeri	3	66.7	0.0	33.3	66.7	100.0
Salinicoccus roseus	3	100.0	33.3	66.7	66.7	66.7
Staphylococcus aureus	6	66.7	33.3	50.0	100.0	83.3
Tenacibaculum aestuarii	1	100.0	100.0	100.0	100.0	100.0
Tenacibaculum lutimaris	6	83.3	16.7	66.7	100.0	100.0
Vibrio alginolyticus	16	62.5	68.8	87.5	93.8	87.5
Vibrio azureus	6	50.0	50.0	83.3	100.0	100.0
Vibrio diazotrophicus	6	100.0	0.0	33.3	66.7	100.0
Vibrio mediterranei	8	87.5	37.5	37.5	87.5	100.0
Vibrio parahaemolyticus	20	70.0	40.0	75.0	70.0	95.0
Vibrio vulnificus	8	75.0	12.5	87.5	62.5	87.5
Virgibacillus pantothenticus	4	50.0	25.0	100.0	75.0	75.0

Number total number of multi-drug isolates

Discussion

Aquatic pathogenic bacteria frequently cause diseases in shrimp farms where dense populations of animals are reared under semi-intensive and intensive systems. Although in modern-day shrimp farming there has been availability of probiotics and other beneficial products as well as better management techniques to prevent microbial infections (Thornber et al. 2020), many aquaculture farmers apply antibiotics to treat bacterial infections (Rakesh et al. 2018). As a result, many of the bacteria which cause infections in aquaculture have acquired resistance to these antimicrobials (Thornber et al. 2020). Furthermore, in some regions of the world (south-east Asian countries), in integrated type of shrimp culture systems, it is a regular practice to use poultry and livestock faecal materials as a manure (Shah et al. 2012). The antibiotic residues in the waste have exerted selection force on the aquatic microorganisms that resulted in rapid development and spread of ARGs between the livestock and aquatic environs (Cabello et al. 2013). Application of antimicrobials at sub-therapeutic levels in shrimp culture has also contributed to the

Table 4 MAR index of heterotrophic MDR bacterial isolates from shrimp farm samples and range

Bacterial species	MAR index			Range of MAR index	
	Average	Min	Max	0.6–0.8	0.8–1
Acinetobacter soli	0.64	0.6	0.8	80.00%	20.00%
Bacillus haikouensis	0.6	0.6	0.6	100.00%	0.00%
Bacillus oryzaecorticis	0.9	0.8	1	0.00%	100.00%
Bacillus paramycoides	0.8	0.8	0.8	0.00%	100.00%
Bacillus safensis	0.7	0.6	1	75.00%	25.00%
Bacillus subtilis	0.64	0.6	0.8	80.00%	20.00%
Bhargavaea beijingensis	0.6	0.6	0.6	100.00%	0.00%
Brachybacterium paraconglomeratum	0.6	0.6	0.6	100.00%	0.00%
Burkholderia cepacia	0.6	0.6	0.6	100.00%	0.00%
Cytobacillus firmus	0.6	0.6	0.6	100.00%	0.00%
Escherichia coli	0.8	0.6	1	50.00%	50.00%
Exiguobacterium mexicanum	0.68	0.6	0.8	60.00%	40.00%
Halobacillus litoralis	0.7	0.6	0.8	50.00%	50.00%
Halomonas ventosae	0.68	0.6	1	66.67%	33.33%
Kocuria flava	0.6	0.6	0.6	100.00%	0.00%
Lysinibacillus sp.	0.6	0.6	0.6	100.00%	0.00%
Mesobacillus foraminis	0.66	0.6	0.8	66.67%	33.33%
Mesobacillus jeotgali	0.66	0.6	0.8	66.67%	33.33%
Mesobacillus subterraneus	0.7	0.6	0.8	50.00%	50.00%
Microbacterium aurum	0.6	0.6	0.6	100.00%	0.00%
Microbacterium koreense	0.6	0.6	0.6	100.00%	0.00%
Pseudoalteromonas piscicida	0.76	0.6	1	60.00%	40.00%
Pseudomonas knackmussii	0.7	0.6	0.8	50.00%	50.00%
Pseudomonas stutzeri	0.6	0.6	0.6	100.00%	0.00%
Salinicoccus roseus	0.66	0.6	0.8	66.67%	33.33%
Staphylococcus aureus	0.66	0.6	0.8	66.67%	33.33%
Tenacibaculum aestuarii	0.8	0.8	0.8	0.00%	100.00%
Tenacibaculum lutimaris	0.72	0.6	1	60.00%	40.00%
Vibrio alginolyticus	0.675	0.6	1	68.75%	31.25%
Vibrio azureus	0.766	0.6	1	33.33%	66.67%
Vibrio diazotrophicus	0.6	0.6	0.6	100.00%	0.00%
Vibrio mediterranei	0.7	0.6	1	75.00%	25.00%
Vibrio parahaemolyticus	0.68	0.6	1	75.00%	25.00%
Vibrio vulnificus	0.65	0.6	0.8	75.00%	25.00%
Virgibacillus pantothenticus	0.65	0.6	0.8	75.00%	25.00%

acquisition of multi-drug resistance in bacteria (Cabello et al. 2013). Similarly, occurrence of MDR bacteria has also been seen in *P. vannamei* culture samples.

Gram negative bacteria, especially *Vibrio* sp. harboured highest MDR isolates, as *Vibrio* species were the most dominant microflora in shrimp culture environments in the present study and also earlier studies (Zhang et al. 2011; Dhayanath et al. 2019; Narayanan et al. 2020). Among the MDR bacterial isolates, maximum resistance was detected for oxytetracycline trailed by other antibiotics (erythromycin, co-trimoxazole, ciprofloxacin and chloramphenicol) which is on par with the previous studies (Zhang et al. 2011; Gao et al. 2012; Kathleen et al. 2016; Liu et al. 2019; Nadella et al. 2021). In the aquaculture settings, tetracyclines and sulphonamides are the maximum and commonly administered antimicrobials (Gao et al. 2012), as they are available at an affordable price (Prasad and Ravishankar 2018). The prevalence of MDR in bacteria against these two classes of antimicrobials is a main concern in shrimp farming in Andhra Pradesh, India. In addition, antibiotics such as tetracyclines, chloramphenicol, ciprofloxacin and erythromycin have been used for treatment of diseases encountered in humans and rearing veterinary animals (Thuy et al. 2011), and the resistance against these antibiotics in *P. vannamei* culture environs is quite alarming.

Table 5 Heterotrophic bacterial populations harbouring different ARGs, MGEs and their prevalence

Bacterial isolates	Antibiotic resistant genes (ARGs)															Mobile Genetic Elements (MGEs)				
	n	Tetracyclines					Sulphona-mides		Macrolides		Chloramphenicol			Quinolones						
		tetA	*tetM*	*tetK*	*tetL*	*tetS*	*sul1*	*sul2*	*ermB*	*ermC*	*catI*	*catII*	*florR*	*qnrA*	*qnrD*	*Int1*	*Int2*	*Tn 1721*	*Tn 1545*	*Tn 917*
Acinetobacter soli	5	4	1	0	0	0	2	0	0	1	1	0	0	0	1	3	1	2	0	0
Bacillus haikouensis	2	1	0	0	0	0	1	0	1	0	0	1	1	1	0	2	1	1	1	0
Bacillus oryzaecorticis	2	1	1	0	0	0	0	1	0	1	0	0	1	0	1	1	1	1	0	0
Bacillus paramycoides	2	1	0	1	0	1	1	1	0	1	1	1	0	0	1	2	1	1	0	0
Bacillus safensis	4	2	1	0	1	0	1	0	1	0	1	1	0	1	1	2	1	0	1	0
Bacillus subtilis	5	2	1	1	0	1	1	1	1	0	0	1	1	1	0	2	2	1	1	0
Bhargavaea beijingensis	1	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	0	0	0
Brachybacterium paraconglomeratum	2	1	0	1	0	0	1	0	1	0	1	0	0	0	1	1	1	0	1	0
Burkholderia cepacia	3	1	0	0	1	1	0	1	0	1	0	1	1	0	1	1	0	1	0	0
Cytobacillus firmus	1	0	0	1	0	0	0	0	1	0	0	1	0	0	0	1	0	0	1	0
Escherichia coli	4	1	1	0	0	1	1	1	3	2	0	0	1	0	1	2	1	0	2	1
Exiguobacterium mexicanum	5	2	1	1	1	0	2	1	2	1	1	1	1	0	1	3	1	1	1	0
Halobacillus litoralis	4	2	0	1	0	1	1	0	0	1	0	1	1	1	1	1	1	1	0	0
Halomonas ventosae	9	4	2	1	1	1	3	2	2	1	2	1	1	1	1	5	2	1	1	1
Kocuria flava	2	1	0	0	1	0	1	0	0	1	0	1	0	0	1	1	0	0	0	0
Lysinibacillus sp.	2	0	1	0	0	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0
Mesobacillus foraminis	3	1	1	1	0	0	1	1	0	1	1	0	1	0	1	2	1	1	0	0
Mesobacillus jeotgali	3	2	0	1	0	0	1	1	1	0	1	1	0	1	0	1	1	1	1	0
Mesobacillus subterraneus	2	1	0	0	0	1	0	1	1	0	1	0	0	0	1	1	0	0	1	0
Microbacterium aurum	4	2	1	0	1	0	1	1	1	0	0	1	1	1	1	2	1	1	1	0
Microbacterium koreense	2	1	0	1	0	0	1	0	0	1	0	1	0	0	1	1	0	0	0	0
Pseudoalteromonas piscicida	5	3	0	1	1	0	2	1	1	0	1	1	0	1	1	2	1	1	1	1
Pseudomonas knackmussii	2	1	1	0	0	0	1	0	0	1	1	0	1	0	1	0	0	1	0	0
Pseudomonas stutzeri	3	1	0	0	1	1	0	1	0	1	1	1	0	1	0	1	1	0	0	1
Salinicoccus roseus	3	2	0	1	1	0	1	1	1	0	1	0	1	1	1	1	1	1	1	0
Staphylococcus aureus	6	3	1	0	0	1	2	1	1	1	1	1	0	1	1	2	1	1	1	1
Tenacibaculum aestuarii	1	0	0	0	1	0	0	1	0	1	0	0	0	1	0	0	0	0	0	0
Tenacibaculum lutimaris	6	3	1	1	0	1	2	1	1	1	2	1	0	1	1	3	1	0	1	1
Vibrio alginolyticus	15	9	3	1	1	1	3	2	2	2	3	1	2	1	2	7	2	2	1	1
Vibrio azureus	6	2	0	1	0	1	1	1	0	1	1	1	1	1	1	3	1	1	0	0
Vibrio diazotrophicus	6	3	1	1	0	1	2	1	1	2	2	1	0	2	1	2	1	1	1	0
Vibrio mediterranei	8	4	1	0	1	1	2	1	2	1	1	2	2	1	2	4	2	1	1	0

Table 5 (continued)

Bacterial isolates	*n*	Antibiotic resistant genes (ARGs)														Mobile Genetic Elements (MGEs)				
		Tetracyclines					Sulphona-mides		Macrolides		Chloramphenicol			Quinolones						
		tetA	*tetM*	*tetK*	*tetL*	*tetS*	*sul1*	*sul2*	*ermB*	*ermC*	*catI*	*catII*	*florR*	*qnrA*	*qnrD*	*Int1*	*Int2*	*Tn 1721*	*Tn 1545*	*Tn 917*
Vibrio parahaemolyticus	20	10	4	2	2	2	4	2	3	2	3	2	2	3	3	9	3	2	2	1
Vibrio vulnificus	8	4	1	0	1	1	2	1	2	1	2	1	0	2	2	3	2	1	1	0
Virgibacillus pantothenticus	4	1	0	1	0	1	1	0	1	0	1	0	1	0	1	2	0	0	0	0
Prevalence (%)		47.5	15.0	11.9	9.4	11.9	26.3	16.9	18.8	17.5	19.4	15.6	13.1	15.6	20.0	46.2	18.7	15.6	13.7	5

n total number of resistant isolates

Antimicrobial resistance of bacteria might depend on the selection pressure which can change rapidly based on exposure time and varies with the geographical location (Satyanarayana 2009). There has been a wide variation in location-wise prevalence of MDR pattern in this study, and this can be attributed to difference in the source of post-larvae procured from hatcheries and numbers of seed stocked in culture ponds as antimicrobials are commonly applied in hatcheries as prophylactic agents and the resistance developed may be carried forward to the aquaculture ponds (Zhang et al. 2011). Furthermore, the water drained out while harvesting the shrimps and disposal of upper layer of sediment during pond preparation also increase the risk related to antibiotic contamination in the shrimp farming (Lai et al. 2018). The use of commercial feed and feed additives supplemented with antibiotic substances also supports the advent and distribution of AMR (Thuy et al. 2011).

The perusal of resistance pattern indicated that many *Vibrio* sp. (*Vibrio alginolyticus*, *V. azureus*, *V. diazotrophicus*, *V. mediterranei*, *V. parahaemolyticus* and *V. vulnificus*) have recorded higher MDR compared to other heterotrophic bacteria to the tested antibiotics. Similar pattern has been reported in earlier studies from shrimp culture environments that harboured increased drug resistance (Pham et al. 2018; Dhayanath et al. 2019; Narayanan et al. 2020). In this study, we have observed the occurrence of MDR in *Bacillus* sp. which are commonly represented in beneficial microbial probiotics. Recent studies have described and reviewed the incidence of MDR in probiotic bacteria (Li et al. 2020; Anokyewaa et al. 2021). In addition, the feed probiotic products and antimicrobial agents left-over in the bottom of shrimp pond are also responsible for the emergence of MDR in heterotrophic bacterial populations (Thuy et al. 2011). Together with *Vibrio* sp., the prevalence of different antibiotic-resistant human opportunistic pathogenic bacteria such as *Acinetobacter soli*, *Burkholderia cepacia*, *Escherichia coli*, *Pseudomonas knackmussii*, *P. stutzeri* and *Staphylococcus aureus* and aquatic pathogenic bacteria namely *Halomonas ventosae*, *Pseudoalteromonas piscicida*, *Tenacibaculum aestuarii* and *Tenacibaculum lutimaris* have also been noticed in this study. Similarly, the prevalence of MDR in *Acinetobacter*, *Bacillus*, *Burkholderia* sp., and some other bacteria from shrimp ponds have been described (Gao et al. 2012; Capkin et al. 2017; Pham et al. 2018). Besides, detection of multi-drug resistant *Acinetobacter*, *Escherichia coli* and *Pseudomonas* sp. have been reported from farm sediments (Osman et al. 2021). It is known that several opportunistic pathogens such as *Acinetobacter* sp. *Burkholderia* sp. and *Pseudomonas* sp. have the capacity to acquire new drug resistance along with intrinsic resistance to different antibiotics (Popowska et al. 2010).

MAR index can be used as a tool to distinguish the risk associated with the application of antibiotics in a particular

Table 6 Conjugation transfer frequency of MDR bacterial species, ARGs, MGEs detected in transconjugants and their MIC

Isolate	Bacterial species	Conjugation transfer frequency					ARGs and MGEs detected in transconjugants	MIC of transconjugants (µg mL^{-1})				
		CIP	C	COT	E	O		CIP	C	COT	E	O
L1BSh3	*Tenacibaculum lutimaris*	$1{\cdot}1\times10^{-8}$	5.6×10^{-4}	7.2×10^{-3}	2.3×10^{-7}	3.5×10^{-7}	*tetA*, *sul1*, *ermB*, *qnrA int1*, *Tn1721*, *Tn1545*	0.25	64	256	64	64
L2CS9	*Pseudoalteromonas piscicida*	4.6×10^{-8}	4.2×10^{-4}	1.8×10^{-3}	5.1×10^{-7}	3.4×10^{-7}	*qnrA*, *ermB*, *int1*, *Tn1545*	0.25	64	128	32	128
L5ES3	*Halomonas ventosae*	1.9×10^{-8}	5.3×10^{-4}	5.6×10^{-3}	6.1×10^{-7}	6.6×10^{-7}	*sul1*, *ermB*, *qnrA*, *int1*, *Tn1545*	0.25	64	128	64	256
L5CSh9	*Escherichia coli*	1.6×10^{-8}	3.8×10^{-3}	2.9×10^{-5}	4.4×10^{-7}	1.0×10^{-8}	*tetA*, *sul1*, *int1*, *Tn1721*	0.5	128	512	128	512
L5ESh7	*Vibrio parahaemolyticus*	3.1×10^{-8}	6.1×10^{-4}	9.4×10^{-4}	1.1×10^{-9}	1.2×10^{-8}	*tetA*, *ermB*, *qnrA*, *Tn1721*, *Tn1545*	0.25	64	512	64	64
L11ESh10	*Pseudoalteromonas piscicida*	3.4×10^{-7}	4.8×10^{-4}	4.4×10^{-3}	2.3×10^{-8}	2.6×10^{-7}	*int1*, *ermB*, *qnrA*, *Tn1545*	0.25	64	256	32	256
L9ESh2	*Vibrio alginolyticus*	3.9×10^{-7}	4.9×10^{-4}	8.1×10^{-3}	3.9×10^{-8}	6.9×10^{-7}	*tetA*, *sul1*, *intI*, *Tn1721*	0.5	64	128	64	128
L6CS4	*Vibrio mediterranei*	4.5×10^{-7}	4×10^{-4}	5.3×10^{-4}	4.6×10^{-8}	3.1×10^{-7}	*sul1*, *ermB*, *qnrA*, *intI*, *Tn1545*	0.25	64	128	64	128
L6DS8	*Vibrio parahaemolyticus*	6.6×10^{-7}	6.5×10^{-4}	6.6×10^{-4}	6.6×10^{-8}	7.8×10^{-7}	*ermB*, *qnrA*,, *intI*, *Tn1545*	0.25	64	256	128	512
L6DW4	*Bacillus safensis*	1.3×10^{-6}	3.6×10^{-4}	1.4×10^{-3}	5.6×10^{-8}	3.8×10^{-7}	*tetA*, *sul1*, *ermB*, *qnrA*, *intI*, *Tn1721*, *Tn1545*	0.25	32	64	16	32
L6DSh6	*Escherichia coli*	3.2×10^{-8}	3.9×10^{-4}	6.6×10^{-3}	1.6×10^{-8}	9.8×10^{-6}	*ermB*, *qnrA*, *intI*, *Tn1545*	0.5	128	512	128	256
L7BW3	*Vibrio mediterranei*	1.6×10^{-7}	4.4×10^{-4}	4.6×10^{-4}	9.7×10^{-8}	2.2×10^{-7}	*sul1*, *ermB*, *qnrA*, *int1*, *Tn1545*	0.25	64	256	64	128
L7BSh1	*Vibrio parahaemolyticus*	7.9×10^{-7}	6.5×10^{-5}	7.1×10^{-3}	4.7×10^{-8}	4.3×10^{-7}	*tetA*, *sul1*, *qnrA*, *intI*, *Tn1721*	0.25	64	128	64	256
L7BSh6	*Vibrio azureus*	3.8×10^{-7}	4.6×10^{-4}	4.9×10^{-4}	8.7×10^{-8}	7.2×10^{-7}	*sul1*, *ermB*, *int1*, *Tn1545*	0.25	64	256	32	64
L7DS12	*Bacillus oryzaecorticis*	3.2×10^{-6}	3.3×10^{-4}	1.9×10^{-3}	6.7×10^{-8}	9.1×10^{-6}	*tetA*, *sul1*, *ermB*, *qnrA*, *intI*, *Tn1721*, *sul1*	0	32	128	32	64

MDR multi-drug resistant, *ARGs* Antibiotic resistant genes, *MGEs* mobile genetic elements, *MIC* minimum inhibitory concentration, *CIP* ciprofloxacin, *C* chloramphenicol, *COT* co-trimoxazole, *E* erythromycin, *O* oxytetracycline location, and the value of 0.2 is used to separate low risk (<0.2) and high risk (≥ 0.2) (Krumperman 1983). The result obtained from these types of investigations will be useful in studying and analysis of resistance pattern of the bacteria from the risk region (Osman et al. 2021). The observations in this study revealed that bacterial species such as *Bacillus oryzaecorticis*, *Bacillus safensis*, *Escherichia coli*, *Halomonas ventosae*, *Pseudoalteromonas piscicida*, *Tenacibaculum lutimaris*, *Vibrio alginolyticus*, *Vibrio azureus*, *Vibrio mediterranei* and *Vibrio parahaemolyticus* had a maximum MAR index of 1 which indirectly indicate the use of antibiotics during culture practice and subsequent exposure of bacterial species to the antibiotics that resulted in the emergence of MDR. Bacterial isolates from shrimp samples also showed the highest MAR index of ≥ 0.2 which suggests that feed taken up by shrimp was adulterated, and the resistance has been acquired. Similarly, there were wide variations in MAR index of bacterial cultures from different sampling locations that can be attributed due to differences in source of seed (post-larvae) and rearing water. Several studies reported high MAR index in bacterial isolates from shrimp culture environments (Zhang et al. 2011; Kathleen et al. 2016; Pham et al. 2018; Narayanan et al. 2020; Reverter et al. 2020).

The predominant ARGs seen in the MDR bacterial isolates were *tetA* (47.5%) followed by *sul1* (26.3%) and *qnrD* (20%) suggesting that ARGs of tetracycline, sulphonamide and quinolone are extensively disseminated in the shrimp ponds of Andhra Pradesh, India. Several reports are existing on the incidence of MDR and ARGs in MDR bacteria (Gao et al. 2012; Su et al. 2017; Zhao et al. 2018; Pham et al. 2018; Dhayanath et al. 2019; Narayanan et al. 2020; Girijan et al. 2020; Nadella et al. 2021). The prevalence of different ARGs in bacterial cultures from the shrimp ponds mainly depend on the frequency in usage of antibiotics and also on the extent of exposure to a particular antibiotic. The excessive use as well as perseverance of antimicrobial agents for extended time resulted in development and dissemination of resistant genes in shrimp aquaculture settings (Chen et al. 2017). Widespread prevalence of ARGs in shrimp culture systems pose a major risk to human health and environment exerted through different mechanisms (Cabello et al. 2013). It is also evident that ARGs in MDR isolates from shrimp culture ponds have been acquired through HRT via MGEs (plasmids, transposons and integrons) (Wang et al. 2012) and can spread to terrestrial bacteria (Su et al. 2017). *Int1* and *int2* which constitutes for class 1 and 2 integrons have been widely distributed in Gram negative pathogens and were associated with the transfer of ARGs (Domínguez et al. 2019). The prevalence of *int1* and *int2* genes belonging to class 1 and 2 integrons in this study was 46% and 19% respectively which is similar to earlier reports (Jacobs and Chenia 2007; Capkin et al. 2017; Domínguez et al. 2019). Class 1 integrons are common among the drug-resistant bacterial isolates and can be considered as marker for the identification of MDR bacteria (Gillings et al. 2015). Transposon, *Tn1721*, has been noticed in 15.6% isolates and is associated with the transfer of *tetA* gene (Schmidt et al. 2001). Similarly, *Tn1545* and *Tn917* have been observed in 13.7 and 5% of MDR bacterial isolates, respectively and are related to transfer of *ermB* gene (Trieu-Cuot et al. 1990; Shaw and Clewell 1985). The MGEs play a significant part in the transfer of ARGs from pathogenic bacteria to susceptible commensal bacteria which have been commonly observed in aquatic microorganisms (Li et al. 2018). Also, HGT by means of MGEs also determines the emergence of resistance in commensal and susceptible bacterial populations (Schmidt et al. 2001).

MIC is an accurate method for testing the microbial resistance to different antibiotics in comparison to other methods. It is important for evaluating antibiotic resistance because the values directly correlate to the dose of antibiotics required for treating the bacterial diseases (Huang et al. 2015). In this study, MIC values recorded for MDR cultures were similar to reported values of ciprofloxacin (Martínez-Martínez et al. 1998), co-trimoxazole (Domínguez et al. 2019), erythromycin (Giovanetti et al. 2002), oxytetracycline (Shin et al. 2015) and chloramphenicol (Abraham 2016).

Conjugation is one of the mechanisms of HGT in which plasmids can carry the ARGs among drug-resistant bacterial populations (Suhartono and Savin, 2016). In conjugation studies, ARGs have also been detected in the transconjugants suggesting the possible role of MGEs in the transfer of ARGs. Integron such as *Int1* has been associated with transfer of *tet* genes (Jacobs and Chenia 2007), *sul1* gene (Fluit and Schmitz 1999) and *qnrA* gene (Li 2005). Similarly, transposons such as *Tn1721* have been involved in transfer of *tetA* (Chenia and Vietze 2012) and *Tn1545* in transfer of *ermB* (Trieu-Cuot et al. 1990). Different ARGs on resistance (R) plasmids which locate on mobile transposons mediate rapid dissemination of genetic determinants, as R transposons could be termed as “jumping gene” (Bennett 2005). This indicates similar transfer is very much possible in aquatic environments and can pose severe threat to aquatic animal health and to the public. However, unlike in in vitro experiments, the frequency of transfer differs in natural environments as it is greatly influenced by several abiotic (temperature, pH, oxygen, nutrient availability, attachment surfaces) and biotic (microbial antagonistics, competitors, symbiotic organisms) factors (Van Elsas and Bailey 2002).

Conclusion

MDR bacteria were widely prevalent in shrimp aquaculture ponds of Andhra Pradesh, India, and this is a grave hazard to the aquatic animal wellbeing, public and environment. The AMR of MDR bacteria was maximum for oxytetracycline (89%) as it was the preferred drug for use in aquaculture. The bacterial profile of the MDR bacteria in shrimp farms included not only shrimp pathogens (*Vibrio* sp.) but also human pathogens (*Escherichia coli*, *Staphylococcus aureus*, *Burkholderia cepacia*, *Acinetobacter soli*), fish pathogens (*Pseudoalteromonas piscicida*, *Tenacibaculum lutimaris*, *Tenacibaculum aestuarii*, *Pseudomonas stutzeri*, *Pseudomonas knackmussii*, *Halomonas ventosae*) and beneficial bacteria (*Bacillus* sp.). The MDR bacteria also harboured different ARGs that might act as a diverse gene pool for the quick progress and distribution of antibiotic resistance to other commensal bacteria in newer locations. Conjugation studies revealed that MDR isolates were capable of transferring the ARGs between the bacterial species by the means of MGEs such as integrons and transposons, as both were detected in transconjugants. Therefore, it is necessary to have a better knowledge on the molecular mechanisms related to the phenotype and genotype of the MDR bacterial isolates to correctly appraise the risk associated with multi-drug resistance in shrimp aquaculture and strategize mitigation measures under one health approach.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11356-021-18163-1.

Acknowledgements The authors thank Directors of ICAR-CIFE (Central Institute of Fisheries Education), Mumbai and ICAR-CIFT (Central Institute of Fisheries Technology), Cochin, India, for the necessary facilities extended for the present study.

Author contribution R.K.N. involved in concept development, collection of samples, execution of experiments, preparation of manuscript.

S.K.P. involved in concept development, designing of experiments, data interpretation and editing of manuscript.

M.R.B. designing of sampling protocol, collection of samples, and editing of manuscript.

P.P.K. involved in arranging field and lab facilities, editing and performing statistical analysis.

R.R.P. involved in concept development and research supervision.

M.P.M. involved in concept development, designing of sampling protocol, acquisition of institute fund, editing of manuscript and overall supervision of the present study.

Funding This work was supported by Indian Council of Agricultural Research, New Delhi, India.

Data availability The Complete data analysed during this study are available with corresponding author.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

References

Abraham TJ (2016) Transferable chloramphenicol resistance determinant in luminous *Vibrio harveyi* from penaeid shrimp *Penaeus monodon* larvae. J Fish 4(3):428–430

Anokyewaa MA, Amoah K, Li Y, Lu Y, Kuebutornye FK, Asiedu B, Seidu I (2021) Prevalence of virulence genes and antibiotic susceptibility of *Bacillus* used in commercial aquaculture probiotics in China. Aquac Rep 21:100784. https://doi.org/10.1016/j.aqrep.2021.100784

Bennett PM (2005) Genome plasticity. In: Woodford N, Johnson A (Eds). Methods in molecular biology, Vol 266, Genomics, Proteomics and Clinical Bacteriology. Humana Press Inc.: Totowa, NJ, pp. 71–113

Bermúdez-Almada MC, Espinosa-Plascencia A (2012) The use of antibiotics in shrimp farming. In: Carvalho E, Silva G, Silva R (eds) Health and Environment in Aquaculture. Open Access: InTech. Publishing, pp.199–214. https://doi.org/10.5772/28527

Brenner DJ, Krieg NR, Staley JT, Garrity GM, Boone DR, De Vos P, GoodFellow M, Rainey FA, Schleifer K (2005) Bergey's manual of systemic bacteriology volume two: the proteobacteria, Part B: The Gammaproteobacteria. Springer, Boston

Cabello FC, Godfrey HP, Tomova A, Ivanova L, Dolz H, Mil-E-lanao A (2013) Antimicrobial use in aquaculture re-examined: its relevance to antimicrobial resistance and to animal and human health. Environ Microbiol 15:1917–1942. https://doi.org/10.1111/1462-2920.12134

Capkin E, Ozdemir S, Ozturk RC, Altinok I (2017) Determination and transferability of plasmid-mediated antibiotic resistance genes of the bacteria isolated from rainbow trout. Aquac Res 48(11):5561–5575. https://doi.org/10.1111/are.13378

Capkin E, Terzi E, Altinok I (2015) Antibiotic resistance gene occurrence in culturable bacteria isolated from Turkish trout farms and their local aquatic environment. Dis Aquat Org 114:127–137. https://doi.org/10.3354/dao02852

Chen CQ, Zheng L, Zhou JL, Zhao H (2017) Persistence and risk of antibiotic residues and antibiotic resistance genes in major mariculture sites in Southeast China. Sci Total Environ 580:1175–1184. https://doi.org/10.1016/j.scitotenv.2016.12.075

Chenia HY, Vietze C (2012) Tetracycline resistance determinants of heterotrophic bacteria isolated from a South African tilapia aquaculture system. Afr J Microbiol Res 6:6761–6768. https://doi.org/10.5897/AJMR10.840

CLSI (2020) Methods for antimicrobial broth dilution and disk diffusion susceptibility testing of bacteria isolated from aquatic animals. Clinical and Laboratory Standards Institute, 2nd ed. CLSI guideline VET03. Wayne

Dhayanath M, Prasad PK, Juliet A, Mary SJ, Paul T, Majethia H (2019) Prevalence of antimicrobial resistance among *Vibrio* spp. isolated from the digestive tract of cultured *Penaeus vannamei*. J Anim Res 9(5):675–681. https://doi.org/10.30954/2277-940X.05.2019.7

Domínguez M, Miranda CD, Fuentes O, de la Fuente M, Godoy FA, Bello-Toledo H, González-Rocha G (2019) Occurrence of transferable integrons and *sul* and *dfr* genes among sulfonamide-and/or trimethoprim-resistant bacteria isolated from Chilean salmonid farms. Front Microbiol 10:748. https://doi.org/10.3389/fmicb.2019.00748

FAO (2020) The state of world fisheries and aquaculture 2020. Sustainability in action. Food and Agriculture Organization of the United Nations Rome. https://doi.org/10.4060/ca9229en

Gao P, Mao D, Luo Y, Wang L, Xu B, Xu L (2012) Occurrence of sulfonamide and tetracycline-resistant bacteria and resistance genes in aquaculture environment. Water Res 46:2355–2364. https://doi.org/10.1016/j.watres.2012.02.004

Gillings MR, Gaze WH, Pruden A, Smalla K, Tiedje JM, Zhu YG (2015) Using the class 1 integron-integrase gene as a proxy for anthropogenic pollution. ISME J 9:1269–1279. https://doi.org/10.1038/ismej.2014.226

Giovanetti E, Magi G, Brenciani A, Spinaci C, Lupidi R, Facinelli B, Varaldo PE (2002) Conjugative transfer of the *erm* (A) gene from erythromycin-resistant *Streptococcus pyogenes* to macrolide-susceptible *S. pyogenes, Enterococcus faecalis and Listeria innocua*. J Antimicrob Chemother 50(2):249–252. https://doi.org/10.1093/jac/dkf122

Girijan SK, Paul R, Rejish VKJ, Pillai D (2020) Investigating the impact of hospital antibiotic usage on aquatic environment and aquaculture systems: a molecular study of quinolone resistance in *Escherichia coli*. Sci Total Environ 748:141538. https://doi.org/10.1016/j.scitotenv.2020.141538

Guo XP, Liu X, Niu Z, Lu D, Zhao S, Sun X (2018) Seasonal and spatial distribution of antibiotic resistance genes in the sediments along the Yangtze estuary, China. Environ Pollut 242:576–584. https://doi.org/10.1016/j.envpol.2018.06.099

Hong B, Ba Y, Niu L, Lou F, Zhang Z, Liu H, Pan Y, Zhao Y (2018) A comprehensive research on antibiotic resistance genes in microbiota of aquatic animals. Front Microbiol 9:1617. https://doi.org/10.3389/fmicb.2018.01617

Huang Y, Zhang L, Tiu L, Wang HH (2015) Characterization of antibiotic resistance in commensal bacteria from an aquaculture ecosystem. Front Microbiol 6:914. https://doi.org/10.3389/fmicb.2015.00914

Jacobs L, Hafizah Y, Chenia YH (2007) Characterization of integrons and tetracycline resistance determinants in *Aeromonas* spp. isolated from South African aquaculture. Int J Food Microbiol 114:295–306. https://doi.org/10.1016/j.ijfoodmicro.2006.09.030

Kathleen MM, Samuel L, Felecia C, Reagan EL, Kasing A, Lesley M (2016) Antibiotic resistance of diverse bacteria from aquaculture in Borneo. Int J Microbiol 16:2164761. https://doi.org/10.1155/2016/2164761

Krumperman PH (1983) Multiple antibiotic resistance indexing of *Escherichia coli* to identify high-risk sources of fecal contamination of foods. Appl Environ Microbiol 46(1):165–170

Lai WWP, Lin YC, Wang YH, Guo YL, Lin AYC (2018) Occurrence of emerging contaminants in aquaculture waters: cross-contamination between aquaculture systems and surrounding waters. Water Air Soil Pollut 229:249. https://doi.org/10.1007/s11270-018-3901-3

Leungtongkam U, Thummeepak R, Tasanapak K, Sitthisak S (2018) Acquisition and transfer of antibiotic resistance genes in association with conjugative plasmid or class 1 integrons of *Acinetobacter baumannii*. PLoS ONE 13(12):e0208468. https://doi.org/10.1371/journal.pone.0208468

Li L, Dechesne A, He Z, Madsen JS, Nesme J, Sørensen SJ (2018) Estimating the transfer range of plasmids encoding antimicrobial resistance in a wastewater treatment plant microbial community. Env Sci Technol Lett 5:260–265. https://doi.org/10.1021/acs.estlett.8b00105

Li T, Teng D, Mao R, Hao Y, Wang X, Wang J (2020) A critical review of antibiotic resistance in probiotic bacteria. Food Res Int 136:109571. https://doi.org/10.1016/j.foodres.2020.109571

Li XZ (2005) Quinolone resistance in bacteria: emphasis on plasmid mediated mechanisms. Int J Antimicrob Agents 25:453–463. https://doi.org/10.1016/j.ijantimicag.2005.04.002

Liu K, Han J, Li S, Liu L, Lin W, Luo J (2019) Insight into the diversity of antibiotic resistance genes in the intestinal bacteria of shrimp *Penaeus vannamei* by culture-dependent and independent approaches. Ecotoxicol Environ Saf 172:451–459. https://doi.org/10.1016/j.ecoenv.2019.01.109

Magiorakos AP, Srinivasan A, Carey RT, Carmeli Y, Falagas MT, Giske CT, Harbarth S, Hindler JT, Kahlmeter G, Olsson-Liljequist B, Paterson DT (2012) Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. Clin Microbiol Infect 18(3):268–281. https://doi.org/10.1111/j.1469-0691.2011.03570.x

Marti E, Variatza E, Balcazar JL (2014) The role of aquatic ecosystems as reservoirs of antibiotic resistance. Trends Microbiol 22:36–41. https://doi.org/10.1016/j.tim.2013.11.001

Martínez-Martínez L, Pascual A, Jacoby GA (1998) Quinolone resistance from a transferable plasmid. Lancet 351(9105):797–799. https://doi.org/10.1016/S0140-6736(97)07322-4

MPEDA (2020) Annual Report 2019–20. The Marine Products Export Development Authority. Cochin 83

Murugadas V, Rao BM, Prasad MM, Ravishankar CN (2019) Antibiotic resistances: genes and associated mechanisms. In: Antimicrobial resistance. Concepts, methodologies and strategies to overcome (Eds: Sharma SK, Vikas G, Agrawal M, Faridi F). Narendra Publishing House, New Delhi. 148–167 p

Nadella RK, Panda SK, Rao BM, Prasad KP, Raman RP, Mothadaka MP (2021) Antibiotic resistance of culturable heterotrophic bacteria isolated from shrimp (*Penaeus vannamei*) aquaculture ponds. Mar Pollut Bull 172:112887. https://doi.org/10.1016/j.marpolbul.2021.112887

Narayanan SV, Joseph TC, Peeralil S, Koombankallil K, Vaiyapuri M, Mothadaka MP, Lalitha KV (2020) Tropical shrimp aquaculture farms harbour pathogenic *Vibrio parahaemolyticus* with high genetic diversity and Carbapenam resistance. Mar Pollut Bull 160(11):111551. https://doi.org/10.1016/j.marpolbul.2020.111551

Nikaido H (2009) Multidrug resistance in bacteria. Ann Rev Biochem 78:119–146. https://doi.org/10.1146/annurev.biochem.78.082907.145923

Osman KM, da Silva PÁ, Franco OL, Saad A, Hamed M, Naim H, Ali AH, Elbehiry A (2021) Nile tilapia (*Oreochromis niloticus*) as an aquatic vector for *Pseudomonas* species of medical importance: antibiotic resistance association with biofilm formation, quorum sensing and virulence. Aquaculture 532:736068. https://doi.org/10.1016/j.aquaculture.2020.736068

Panda SK, Ravishankar CN (2018) Safety of aquacultured products: emerging issues and immediate challenges. In: Zero hunger in India. Policies and perspectives. (Eds. Peter KV), Brillion publishing, New Delhi pp. 509–518

Pham TTH, Rossi P, Dinh HDK, Pham NTA, Tran PA, Ho TTKM (2018) Analysis of antibiotic multi-resistant bacteria and resistance genes in the effluent of an intensive shrimp farm (Long An, Vietnam). J Environ Manage 214:149–156. https://doi.org/10.1016/j.jenvman.2018.02.089

Popowska M, Miernik A, Rzeczycka M, Lopaciuk A (2010) The impact of environmental contamination with antibiotics on levels of resistance in soil bacteria. J Environ Qual 39:1679–1687. https://doi.org/10.2134/jeq2009.0499

Prasad MM, Ravishankar CN (2018) The antimicrobial resistance (AMR): the ICARCIFT contribution in addressing the problem. [WWW document]. URL. http://drs.cift.res.in/handle/123456789/4155. Accessed 2 Aug 2021

Preena PG, Swaminathan TR, Kumar VJR, Singh ISB (2020) Antimicrobial resistance in aquaculture: a crisis for concern. Biologia 75:1497–1517. https://doi.org/10.2478/s11756-020-00456-4

Qiao M, Ying GG, Singer AC, Zhu YG (2018) Review of antibiotic resistance in China and its environment. Environ Int 110:160–172. https://doi.org/10.1016/j.envint.2017.10.016

Rakesh K, Naik GM, Pinto N, Dharmakar P, Pai M, Anjusha KV (2018) A review on drugs used in shrimp aquaculture. Int J Pure Appl Biosci 6(4):77–86. https://doi.org/10.18782/2320-7051.6296

Rao BM, Prasad MM (2015) Residues of veterinary medicinal products (antibiotics) in shrimp exported from India to the European Union (EU)-Trends in the last decade. Fishing Chimes 34(1):6–11

Reverter M, Sarter S, Caruso D, Avarre JC, Combe M, Pepey E, Pouyaud L, Vega-Heredía S, De Verdal H, Gozlan RE (2020) Aquaculture at the crossroads of global warming and antimicrobial resistance. Nat Commun 11(1):1–8. https://doi.org/10.1038/s41467-020-15735-6

Sanderson H, Fricker C, Brown RS, Majury A, Liss SN (2016) Antibiotic resistance genes as an emerging environmental contaminant. Environ Rev 24(2):205–218. https://doi.org/10.1139/er-2015-0069

Satyanarayana K (2009) Detection of antimicrobial resistance in common gram negative and gram positive bacteria encountered in infectious disease-an update. ICMR Res Info Bull 3:1–3

Schmidt AS, Bruun MS, Dalsgaard I, Larsen JL (2001) Incidence, distribution and spread of tetracycline resistance determinants and integron associated antibiotic resistance genes among motile aeromonads from a fish farming environment. Appl Environ Microbiol 67:5675–5682. https://doi.org/10.1128/AEM.67.12.5675-5682.2001

Shah SQA, Colquhoun DJ, Nikuli HL, Sørum H (2012) Prevalence of antibiotic resistance genes in the bacterial flora of integrated fish farming environments of Pakistan and Tanzania. Environ Sci Technol 46:8672–8679. https://doi.org/10.1021/es3018607

Sharma VK, Johnson N, Cizmas L, McDonald TJ, Kim H (2016) A review of the influence of treatment strategies on antibiotic resistant bacteria and antibiotic resistance genes. Chemosphere 150:702–714. https://doi.org/10.1016/j.chemosphere.2015.12.084

Shaw JH, Clewell DB (1985) Complete nucleotide sequence of macrolide-lincosamide-streptogramin B resistance transposon Tn917 in *Streptococcus faecalis*. J Bacteriol 164:782–796. https://doi.org/10.1128/jb.164.2.782-796.1985

Shin SW, Shin MK, Jung M, Belaynehe KM, Yoo HS (2015) Prevalence of antimicrobial resistance and transfer of tetracycline resistance genes in *Escherichia coli* isolates from beef cattle. Appl Environ Microbiol 81:5560–5566. https://doi.org/10.1128/AEM.01511-15

Su H, Liu S, Hu X, Xu X, Xu W, Xu Y, Li Z, Wen G, Liu Y, Cao Y (2017) Occurrence and temporal variation of antibiotic resistance genes (ARGs) in shrimp aquaculture: ARGs dissemination from farming source to reared organisms. Sci Total Environ 607:357–366. https://doi.org/10.1016/j.scitotenv.2017.07.040

Suhartono S, Savin M (2016) Conjugative transmission of antibiotic-resistance from stream water *Escherichia coli* as related to number of sulfamethoxazole but not class 1 and 2 integrase genes. Mob Genet Elements 6(6):e1256851. https://doi.org/10.1080/2159256X.2016.1256851

Teng T, Liang L, Chen K, Xi B, Xie J, Xu P (2017) Isolation, identification and phenotypic and molecular characterization of pathogenic *Vibrio vulnificus* isolated from *Litopenaeus vannamei*. PLoS ONE 12(10):e0186135. https://doi.org/10.1371/journal.pone.0186135

Thornber K, Verner-Jeffreys D, Hinchliffe S, Rahman MM, Bass D, Tyler CR (2020) Evaluating antimicrobial resistance in the global shrimp industry. Rev Aquac 12(2):966–986. https://doi.org/10.1111/raq.12367

Thuy HTT, Nga LP, Loan TTC (2011) Antibiotic contaminants in coastal wetlands from Vietnamese shrimp farming. Environ Sci Pollut Res 18:835–841. https://doi.org/10.1007/s11356-011-0475-7

Trieu-Cuot P, Poyart-Salmeron C, Carlier C, Courvalin P (1990) Nucleotide sequence of the erythromycin resistance gene of the conjugative transposon *Tn1545*. Nucleic Acids Res 18:3660. https://doi.org/10.1093/nar/18.12.3660

Van Elsas JD, Bailey MJ (2002) The ecology of transfer of mobile genetic elements. FEMS Microbiol Ecol 42(2):187–197. https://doi.org/10.1111/j.1574-6941.2002.tb01008.x

Von Wintersdorff CJ, Penders J, Van Niekerk JM, Mills ND, Majumder S, Van Alphen LB, Savelkoul PH, Wolffs PF (2016) Dissemination of antimicrobial resistance in microbial ecosystems through horizontal gene transfer. Front Microbiol 7:173. https://doi.org/10.3389/fmicb.2016.00173

Wang H, Mcentire JC, Zhang L, Li X, Doyle M (2012) The transfer of antibiotic resistance from food to humans: facts, implications and future directions. Rev Sci Tech 31:249–260. https://doi.org/10.20506/rst.31.1.2117

Weisburg WG, Barns SM, Pelletier DA, Lane DJ (1991) 16S ribosomal DNA amplification for phylogenetic study. J Bacteriol 173:697–703

Zhang S, Abbas M, Rehman MU, Huang Y, Zhou R, Gong S, Yang H, Chen S, Wang M, Cheng A (2020) Dissemination of antimicrobial resistance genes (ARGs) via integrons in *Escherichia coli*: a risk to human health. Environ Pollut 226(2):115260. https://doi.org/10.1016/j.envpol.2020.115260

Zhang YB, Li Y, Sun XL (2011) Antibiotic resistance of bacteria isolated from shrimp hatcheries and cultural ponds on Donghai Island, China. Mar Pollut Bull 62:2299–2307. https://doi.org/10.1016/j.marpolbul.2011.08.048

Zhao Y, Zhang XX, Zhao Z, Duan C, Chen H, Wang M, Ren H, Yin Y, Ye L (2018) Metagenomic analysis revealed the prevalence of antibiotic resistance genes in the gut and living environment of freshwater shrimp. J Hazard Mater 350:10–18. https://doi.org/10.1016/j.jhazmat.2018.02.004

Zhou R, Zeng S, Hou D, Liu J, Weng S, He J, Huang Z (2019) Occurrence of human pathogenic bacteria carrying antibiotic resistance genes revealed by metagenomic approach: a case study from an aquatic environment. J Environ Sci 80:248–256. https://doi.org/10.1016/j.jes.2019.01.001

Zhou R, Zeng S, Hou D, Liu J, Weng S, He J, Huang Z (2020) Temporal variation of antibiotic resistance genes carried by culturable bacteria in the shrimp hepatopancreas and shrimp culture pond water. Ecotoxicol Environ Saf 199:110738. https://doi.org/10.1016/j.ecoenv.2020.110738